SYS 3060 A03
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Q1. Modeling a Student's Workout Routine

(a) State Space

Since the student's next activity depends on both their current activity and health status, the state needs to track both. Not every (activity, health) pair is reachable: upper-body workouts (A, B, C) require the student to be healthy. So the state space is

$$S = \{(A, H),\ (B, H),\ (C, H),\ (L, H),\ (R, H),\ (L, I),\ (R, I)\},$$

giving 7 states total.

(b) Transition Matrix

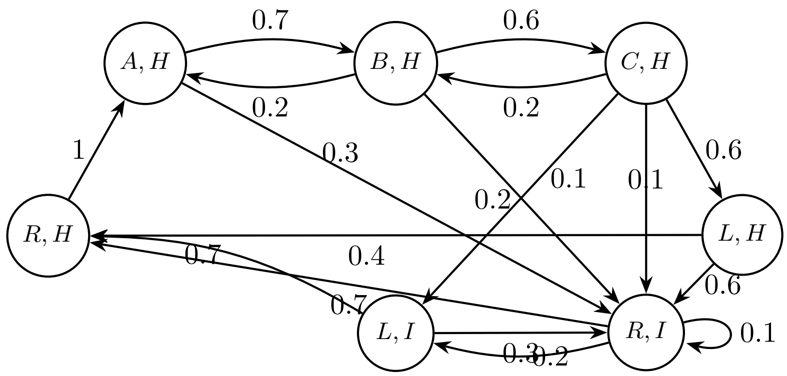
Reading off each transition rule:

- (A, H): back healthy w.p. 0.7, otherwise injured/rest $\Rightarrow (B, H) = 0.7$, $(R, I) = 0.3$.
- (B, H): chest w.p. 0.6, arms w.p. 0.2, otherwise injured/rest $\Rightarrow (C, H) = 0.6$, $(A, H) = 0.2$, $(R, I) = 0.2$.
- (C, H): legs w.p. 0.6, back w.p. 0.2; the remaining 0.2 splits equally between rest-injured and legs-injured $\Rightarrow (L, H) = 0.6$, $(B, H) = 0.2$, $(R, I) = 0.1$, $(L, I) = 0.1$.
- (L, H): rest healthy w.p. 0.4, otherwise injured/rest $\Rightarrow (R, H) = 0.4$, $(R, I) = 0.6$.
- (R, H): arms w.p. 1 $\Rightarrow (A, H) = 1$.
- (L, I): rest injured w.p. 0.3, otherwise recover/rest healthy $\Rightarrow (R, I) = 0.3$, $(R, H) = 0.7$.
- (R, I): rest injured w.p. 0.1, legs injured w.p. 0.2, otherwise recover/rest healthy $\Rightarrow (R, I) = 0.1$, $(L, I) = 0.2$, $(R, H) = 0.7$.

Ordering the states as $(A, H), (B, H), (C, H), (L, H), (R, H), (L, I), (R, I)$:

$$P = \begin{pmatrix} 0 & 0.7 & 0 & 0 & 0 & 0 & 0.3 \\ 0.2 & 0 & 0.6 & 0 & 0 & 0 & 0.2 \\ 0 & 0.2 & 0 & 0.6 & 0 & 0.1 & 0.1 \\ 0 & 0 & 0 & 0 & 0.4 & 0 & 0.6 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0.7 & 0 & 0.3 \\ 0 & 0 & 0 & 0 & 0.7 & 0.2 & 0.1 \end{pmatrix}.$$

The state transition diagram is shown below.



(c) P(injured and resting after 2 days | start at (L,H))

I need the $((L,H),(R,I))$ entry of P^2. The only intermediate states reachable from (L,H) are (R,H) and (R,I), so:

$$(P^2)_{(L,H),(R,I)} = P_{(L,H),(R,H)} \cdot P_{(R,H),(R,I)} + P_{(L,H),(R,I)} \cdot P_{(R,I),(R,I)} = (0.4)(0) + (0.6)(0.1) = \boxed{0.06}.$$

(d) Starting equally likely from (B,H) or (C,H), find P(at (R,H) after 3 days)

The initial distribution is $\boldsymbol{\alpha} = (0,\ 0.5,\ 0.5,\ 0,\ 0,\ 0,\ 0)$. I need the (R,H) component of $\boldsymbol{\alpha}P^3$, computed iteratively.

$\boldsymbol{\alpha}P$: averaging rows 2 and 3 of P:

$$\mathbf{v}^{(1)} = (0.1,\ 0.1,\ 0.3,\ 0.3,\ 0,\ 0.05,\ 0.15).$$

$\mathbf{v}^{(2)} = \mathbf{v}^{(1)}P$: multiplying through,

$$\mathbf{v}^{(2)} = (0.02,\ 0.13,\ 0.06,\ 0.18,\ 0.26,\ 0.06,\ 0.29).$$

$\mathbf{v}^{(3)} = \mathbf{v}^{(2)}P$: I only need the fifth component (the (R,H) column). The nonzero contributions come from states that transition into (R,H), namely (L,H), (L,I), and (R,I):

$$v_5^{(3)} = 0.18(0.4) + 0.06(0.7) + 0.29(0.7) = 0.072 + 0.042 + 0.203 = \boxed{0.317}.$$

(e) Steady-State Probabilities

Solving $\boldsymbol{\pi}P = \boldsymbol{\pi}$ with $\sum \pi_i = 1$. Writing the balance equations and expressing everything in terms of $\pi_2 \equiv \pi_{(B,H)}$:

From the equations I get the chain of dependencies:

$$\begin{aligned}
\pi_3 &= 0.6\,\pi_2, \\
\pi_4 &= 0.6\,\pi_3 = 0.36\,\pi_2, \\
\pi_2 &= 0.7\,\pi_1 + 0.12\,\pi_2 \;\Rightarrow\; \pi_1 = \tfrac{44}{35}\,\pi_2, \\
\pi_5 &= \pi_1 - 0.2\,\pi_2 = \tfrac{37}{35}\,\pi_2.
\end{aligned}$$

For the injured states, substituting into the remaining balance equations and solving the resulting 2×2 system yields $\pi_7 \approx 1.0371\,\pi_2$ and $\pi_6 \approx 0.2674\,\pi_2$.

Normalizing $\sum \pi_i = 1$ with all terms proportional to π_2:

$$\left(\tfrac{44}{35} + 1 + 0.6 + 0.36 + \tfrac{37}{35} + 0.2674 + 1.0371\right)\pi_2 = 5.5788\,\pi_2 = 1,$$

so $\pi_2 = 0.1793$. The full steady-state distribution is:

$$\begin{aligned}
&\pi_{(A,H)} \approx 0.2254, \quad \pi_{(B,H)} \approx 0.1793, \quad \pi_{(C,H)} \approx 0.1076,\\
&\pi_{(L,H)} \approx 0.0645, \quad \pi_{(R,H)} \approx 0.1895, \quad \pi_{(L,I)} \approx 0.0479,\\
&\pi_{(R,I)} \approx 0.1859.
\end{aligned}$$

The long-run proportion of time spent injured is

$$\pi_{(L,I)} + \pi_{(R,I)} \approx 0.0479 + 0.1859 = \boxed{0.2338}.$$

(f) Expected Number of Days Until First Injury from a Healthy State

Let m_i be the expected hitting time to the set of injured states, starting from healthy state i. Transitions to an injured state contribute 0 remaining time:

$$\begin{aligned}
m_{(A,H)} &= 1 + 0.7\,m_{(B,H)},\\
m_{(B,H)} &= 1 + 0.2\,m_{(A,H)} + 0.6\,m_{(C,H)},\\
m_{(C,H)} &= 1 + 0.2\,m_{(B,H)} + 0.6\,m_{(L,H)},\\
m_{(L,H)} &= 1 + 0.4\,m_{(R,H)},\\
m_{(R,H)} &= 1 + m_{(A,H)}.
\end{aligned}$$

Back-substituting from the bottom up: $m_{(R,H)} = 1 + m_{(A,H)}$, then $m_{(L,H)} = 1.4 + 0.4\,m_{(A,H)}$, then using $m_{(A,H)} = 1 + 0.7\,m_{(B,H)}$ to express $m_{(C,H)}$ in terms of $m_{(B,H)}$ alone, and finally plugging into the equation for $m_{(B,H)}$:

$$m_{(B,H)} = 2.448 + 0.3608\,m_{(B,H)} \implies m_{(B,H)} = \frac{2.448}{0.6392} \approx 3.830.$$

The remaining values follow:

$$\begin{aligned}
&m_{(A,H)} \approx 3.681, \quad m_{(B,H)} \approx 3.830,\\
&m_{(C,H)} \approx 3.490, \quad m_{(L,H)} \approx 2.872,\\
&m_{(R,H)} \approx 4.681.
\end{aligned}$$

If we want a single summary weighted by the healthy-state steady-state distribution:

$$\bar{m} = \frac{\sum_{i \in H} \pi_i\, m_i}{\sum_{i \in H} \pi_i} \approx 3.81 \text{ days}.$$

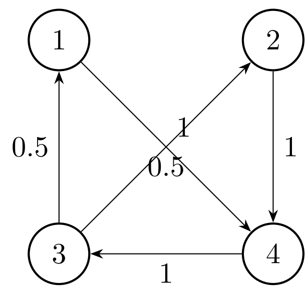
(g) Communicating Classes and Irreducibility

From any state the chain can eventually reach any other: injured states recover to (R, H), which goes to (A, H), which feeds into the workout cycle, and every healthy workout carries some probability of injury. So there is a single communicating class equal to S, and the chain is **irreducible**.

Q2. Markov Chain Classification

(a) P_1

$$P_1 = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 0.5 & 0.5 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$



Every state can reach every other, so the chain is **irreducible** with a single **recurrent** class $\{1, 2, 3, 4\}$. No absorbing states. The chain cycles deterministically as $\{1, 2\} \to \{4\} \to \{3\} \to \{1, 2\}$, giving a partition into three groups that transitions always respect, so the **period is** $d = 3$.

Steady-state probabilities. Solving $\boldsymbol{\pi} P = \boldsymbol{\pi}$:

$$\pi_1 = 0.5\,\pi_3, \quad \pi_2 = 0.5\,\pi_3, \quad \pi_3 = \pi_4, \quad \pi_4 = \pi_1 + \pi_2 = \pi_3.$$

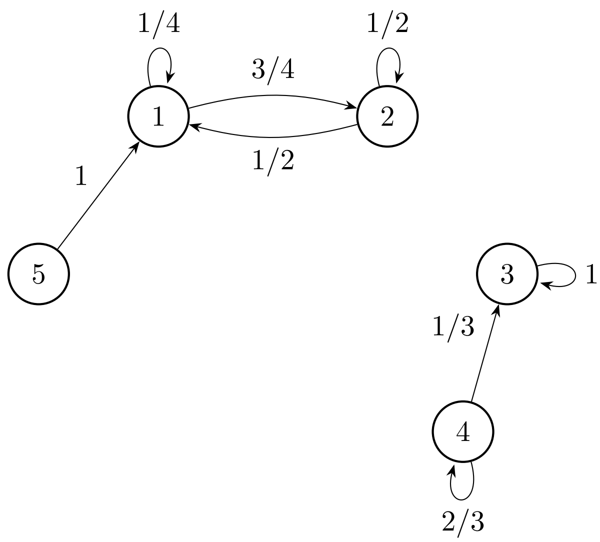
So $\pi_1 = \pi_2$ and $\pi_3 = \pi_4 = 2\pi_1$. Normalizing: $6\pi_1 = 1$, giving

$$\boldsymbol{\pi} = \left(\frac{1}{6}, \frac{1}{6}, \frac{1}{3}, \frac{1}{3}\right).$$

Since $d = 3$, these are time-average proportions; the chain cycles rather than converging pointwise.

(b) P_2

$$P_2 = \begin{pmatrix} 1/4 & 3/4 & 0 & 0 & 0 \\ 1/2 & 1/2 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1/3 & 2/3 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{pmatrix}$$



Classes:

- $\{1, 2\}$: closed, **recurrent**, aperiodic (self-loops give period 1).
- $\{3\}$: **absorbing state**, recurrent, aperiodic.
- $\{4\}$: can reach 3 but not vice versa, so **transient**. Aperiodic ($P_{44} > 0$).
- $\{5\}$: goes to 1 and never returns, so **transient**.

The chain is **not irreducible**.

Steady-state probabilities. Since there are two closed recurrent classes, the steady-state depends on the starting state. For the class $\{1, 2\}$, solving $\pi_1 = \frac{1}{4}\pi_1 + \frac{1}{2}\pi_2$ with $\pi_1 + \pi_2 = 1$ gives $\pi_1 = 2/5$, $\pi_2 = 3/5$.

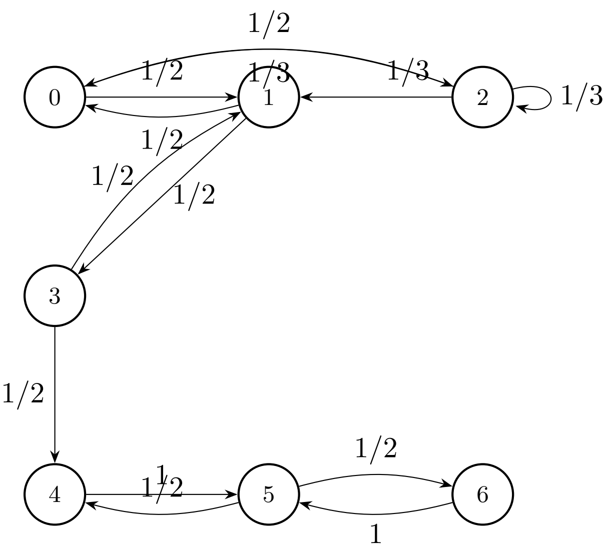
Starting from state 5 or $\{1, 2\}$: $\boldsymbol{\pi} = (2/5,\ 3/5,\ 0,\ 0,\ 0)$.

Starting from state 3 or 4: $\boldsymbol{\pi} = (0,\ 0,\ 1,\ 0,\ 0)$ (state 4 is eventually absorbed into 3 with probability 1).

In general, the stationary distribution is any convex combination of $(2/5, 3/5, 0, 0, 0)$ and $(0, 0, 1, 0, 0)$.

(c) P_3

$$P_3 = \begin{pmatrix} 0 & 1/2 & 1/2 & 0 & 0 & 0 & 0 \\ 1/2 & 0 & 0 & 1/2 & 0 & 0 & 0 \\ 1/3 & 1/3 & 1/3 & 0 & 0 & 0 & 0 \\ 0 & 1/2 & 0 & 0 & 1/2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1/2 & 0 & 1/2 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}$$



Classes:

- $\{0, 1, 2, 3\}$: these states communicate, but state 3 can transition to state 4, so the class is not closed. Therefore these states are **transient**. Aperiodic (state 2 has a self-loop).
- $\{4, 5, 6\}$: closed communicating class, **recurrent**. Returns to state 4 occur at steps $2, 4, 6, \ldots$ (e.g. $4 \to 5 \to 4$), so the **period is** $d = 2$.

The chain is **not irreducible**. No absorbing states.

Long-run fraction of time in state 0: Since state 0 is transient, the chain is eventually absorbed into $\{4, 5, 6\}$ and never returns. The fraction of time in state 0 in the long run is $\boxed{0}$.

Q3. Historical States and Markov Reformulation

(a) Why $\{F, L, B\}$ is Insufficient

If the current study mode is F:

(i) Previous evening was F: the rule $FF \to L$ has probability 0.7, so $P(L \mid FF) = 0.7$.
(ii) Previous evening was L: the rule $LF \to F$ (0.75), $LF \to B$ (0.25) gives $P(L \mid LF) = 0$.

The probability of transitioning to L from state F depends on the previous evening. This violates the Markov property for the state space $\{F, L, B\}$—the process has memory of two steps, not one.

(b) Augmented State Space

We define the state as the ordered pair of the last two evenings' study modes. With three possible modes, the state space has $3^2 = 9$ states:

$$S = \{FF,\ FL,\ FB,\ LF,\ LL,\ LB,\ BF,\ BL,\ BB\}.$$

(c) Transition Matrix

A state (a, b) can only transition to a state of the form (b, c), where c is determined by the transition rules. Reading off each rule:

From	To
FF	FL (0.7), FB (0.3)
FL	LL (0.8), LB (0.2)
FB	BL (0.3), BB (0.7)
LF	FF (0.75), FB (0.25)
LL	LF (0.6), LB (0.4)
LB	BF (0.4), BB (0.6)
BF	FF (0.65), FL (0.35)
BL	LF (0.3), LL (0.7)
BB	BF (0.5), BL (0.5)

In matrix form (rows/columns ordered $FF, FL, FB, LF, LL, LB, BF, BL, BB$):

$$P = \begin{pmatrix} 0 & 0.7 & 0.3 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0.8 & 0.2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0.3 & 0.7 \\ 0.75 & 0 & 0.25 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.6 & 0 & 0.4 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0.4 & 0 & 0.6 \\ 0.65 & 0.35 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.3 & 0.7 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0.5 & 0.5 & 0 \end{pmatrix}.$$

(d) P(state LB two steps after starting in FL)

The first two evenings are F, L, so the initial state is FL.

Step 1: From FL, the chain goes to LL (w.p. 0.8) or LB (w.p. 0.2).

Step 2: From LL: to LF (0.6) or LB (0.4). From LB: to BF (0.4) or BB (0.6).

So $P^2(FL \to LB) = 0.8 \times 0.4 + 0.2 \times 0 = \boxed{0.32}$.

Q4. Inventory Control with Reorder Policies

The demand distribution is $P(D = 0) = 0.2$, $P(D = 1) = 0.5$, $P(D = 2) = 0.3$. Capacity is 3 units, unmet demand is lost, and the state is beginning-of-day inventory.

(a) Policy A: If Ending Inventory ≤ 1, Refill to 3

Any ending inventory of 0 or 1 is refilled to 3 overnight. So the beginning-of-day inventory is either 2 (if ending inventory was 2) or 3 (if ending inventory was ≤ 1 and refilled, or was already 3).

State space: $S_A = \{2, 3\}$.

Tracing the transitions:

From state 3: $D = 0 \Rightarrow$ end 3, next 3. $D = 1 \Rightarrow$ end 2, next 2. $D = 2 \Rightarrow$ end 1, refill to 3, next 3.

From state 2: $D = 0 \Rightarrow$ end 2, next 2. $D = 1 \Rightarrow$ end 1, refill to 3, next 3. $D = 2 \Rightarrow$ end 0, refill to 3, next 3.

$$P_A = \begin{array}{c c} & \begin{array}{cc} 2 & 3 \end{array} \\ \begin{array}{c} 2 \\ 3 \end{array} & \begin{pmatrix} 0.2 & 0.8 \\ 0.5 & 0.5 \end{pmatrix} \end{array}.$$

(b) Policy B: If Ending Inventory $= 0$, Refill to 2

Only ending inventory of exactly 0 triggers an order (refill to 2).

From state 2: $D = 0 \Rightarrow$ end 2, next 2. $D = 1 \Rightarrow$ end 1, no order, next 1. $D = 2 \Rightarrow$ end 0, refill to 2, next 2.

From state 1: $D = 0 \Rightarrow$ end 1, next 1. $D = 1 \Rightarrow$ end 0, refill to 2, next 2. $D = 2 \Rightarrow$ sell 1 (1 lost), end 0, refill to 2, next 2.

State space: $S_B = \{1, 2\}$.

$$P_B = \begin{array}{c c} & \begin{array}{cc} 1 & 2 \end{array} \\ \begin{array}{c} 1 \\ 2 \end{array} & \begin{pmatrix} 0.2 & 0.8 \\ 0.5 & 0.5 \end{pmatrix} \end{array}.$$

(c) Steady-State Distributions

Both matrices have the same structure. Solving $0.8\pi_{\text{low}} = 0.5\pi_{\text{high}}$ with $\pi_{\text{low}} + \pi_{\text{high}} = 1$ gives $\pi_{\text{high}} = 1.6\,\pi_{\text{low}}$, so $2.6\,\pi_{\text{low}} = 1$.

Policy A: $\pi_2 = 5/13 \approx 0.3846, \quad \pi_3 = 8/13 \approx 0.6154.$

Policy B: $\pi_1 = 5/13 \approx 0.3846, \quad \pi_2 = 8/13 \approx 0.6154.$

(d) Long-Run Profit Comparison

I compute the expected daily profit for each starting inventory level. The cost structure is: revenue \$120/unit sold, purchase cost \$70/unit ordered, holding cost \$8/unit of ending inventory (before reorder), fixed ordering cost \$15 per order placed, lost-sale penalty \$25/unit unmet.

Policy A, state 3:

- $D = 0$ (0.2): sell 0, end 3, no order. Profit $= 0 - 24 = -24$.
- $D = 1$ (0.5): sell 1, end 2, no order. Profit $= 120 - 16 = 104$.
- $D = 2$ (0.3): sell 2, end 1, order 2. Profit $= 240 - 140 - 15 - 8 = 77$.

$E[\text{profit} \mid 3] = 0.2(-24) + 0.5(104) + 0.3(77) = 70.3.$

Policy A, state 2:

- $D = 0$ (0.2): sell 0, end 2, no order. Profit $= -16$.
- $D = 1$ (0.5): sell 1, end 1, order 2. Profit $= 120 - 140 - 15 - 8 = -43$.
- $D = 2$ (0.3): sell 2, end 0, order 3. Profit $= 240 - 210 - 15 - 0 = 15$.

$E[\text{profit} \mid 2] = 0.2(-16) + 0.5(-43) + 0.3(15) = -20.2.$

$$E[\text{profit}_A] = \tfrac{5}{13}(-20.2) + \tfrac{8}{13}(70.3) = \frac{-101 + 562.4}{13} \approx \$35.49/\text{day}.$$

Policy B, state 2:

- $D = 0$ (0.2): sell 0, end 2, no order. Profit $= -16$.
- $D = 1$ (0.5): sell 1, end 1, no order. Profit $= 120 - 8 = 112$.
- $D = 2$ (0.3): sell 2, end 0, order 2. Profit $= 240 - 140 - 15 = 85$.

$E[\text{profit} \mid 2] = 0.2(-16) + 0.5(112) + 0.3(85) = 78.3$.

Policy B, state 1:

- $D = 0$ (0.2): sell 0, end 1, no order. Profit $= -8$.
- $D = 1$ (0.5): sell 1, end 0, order 2. Profit $= 120 - 140 - 15 = -35$.
- $D = 2$ (0.3): sell 1 (1 lost), end 0, order 2. Profit $= 120 - 140 - 15 - 25 = -60$.

$E[\text{profit} \mid 1] = 0.2(-8) + 0.5(-35) + 0.3(-60) = -37.1$.

$$E[\text{profit}_B] = \tfrac{5}{13}(-37.1) + \tfrac{8}{13}(78.3) = \frac{-185.5 + 626.4}{13} \approx \$33.92/\text{day}.$$

Conclusion: Policy A produces higher long-run profit (\$35.49/day vs. \$33.92/day). This makes sense: Policy A maintains higher inventory levels, which reduces lost sales and their penalties at relatively modest additional holding and ordering cost.